Cross-Country Relative Value in Government Bond Futures

Alistair J. Chopping

Abstract

We present a systematic rates trading strategy for 10-year government bond futures across a small set of major developed markets. The strategy seeks to exploit relative mispricings in short-horizon, risk-adjusted returns using economically motivated features such as momentum and US Treasury curve structure. We construct a panel of volatility-scaled forward-looking returns and perform a rolling ridge regression to forecast relative performance across a universe of six government bond futures (US, UK, Germany, Canada, France, and Italy). This forecast is then used to construct a relative value portfolio. The strategy demonstrates some degree of predictive power as measured by the information coefficient, though the signal is noisy for reasons we discuss. The strategy shown to be orthogonal to basic momentum, and the residual returns above momentum are positive before transaction costs. However, we find that the signal is sensitive to costs - the residual returns are fully eroded by trading costs in a naive execution strategy. We discuss limitations of our work, including the small cross-sectional universe, the absence of explicit futures roll adjustment, and the need for a more sophisticated execution strategy.

Contents

1 Introduction **3**

2 Strategy Overview **4**

3 Data & Feature Engineering **6**

3.1 Data 6

3.2 Return and Volatility Definitions 6

3.3 Target Construction 6

3.4 Feature Set 7

3.5 Panel Construction 8

4 Modelling Approach **8**

5 Strategy Performance **10**

6 Discussion **16**

6.1 Short Summary 16

6.2 Directions For Future Work 16

A Regression Methods **17**

A.1 OLS Regression 17

A.2 Ridge Regression 19

1 Introduction

This project develops a systematic cross-sectional relative value strategy on 10-year government bond futures across six major developed markets - the United States, United Kingdom, Germany, Canada, France, and Italy. The central research hypothesis is that relative performance across global bond markets is partly driven by momentum effects, changes in interest rate volatility regime, and structure of the US treasury curve, leading to predictable mispricings arising from these macro signals being incorporated into prices at different speeds across different markets.

The strategy constructs a relative value portfolio, ranking assets by forecasting risk-adjusted returns and constructing a portfolio of long positions in the best-performing assets and short positions in the worst. Such a portfolio can generate positive returns irrespective of wider market direction, exploiting mispricings between assets rather than attempting to forecast the direction of the individual assets themselves.

The features used in the model are constructed with economic motivation in mind, and forecasts at a horizon of $h = 25$ days ahead are generated using a panel ridge regression, stabilising coefficient estimates in the presence of likely multicollinearity among the features. The features consist of momentum, yield curve spread and fly, changes in the MOVE index and interaction terms between momentum and curve shape. Momentum is chosen in an effort to capture persistent trends in bond futures, reflecting slow information dissemination and gradual investor rebalancing. US treasury curve spread & fly encode the market's expectations of growth, inflation, and monetary policy. Changes in the MOVE index are used as a proxy for changes in interest rate volatility regime, with volatility and returns traditionally being inversely correlated - a similar effect in equities markets is commonly referred to as the *leverage effect*. The interaction features between curve shape and momentum attempt to capture conditional effects, such as increased or decreased trending behaviour in different macro regimes and market sentiments.

Model performance is evaluated via the Pearson information coefficient, cumulative returns, Sharpe ratios, turnover, and sensitivity to transaction costs. We also show that the strategy's returns are orthogonal to basic momentum.

While we find that the model may have some predictive power and that it generates positive residual returns above momentum, a weak signal coupled with high turnover and naive daily rebalancing significantly erode these residual returns. We discuss potential improvements to the model, paying particular attention to reducing turnover and more realistic cost modelling.

2 Strategy Overview

The central research hypothesis is the following:

> Relative performance across global bond markets is partly driven by momentum effects, changes in interest-rate volatility regime, and US Treasury curve shape, under the hypothesis that such macro signals are incorporated into prices at different speeds across different markets, leading to predictable cross-asset mispricings that can be exploited.

Our goal is to estimate expected volatility-scaled forward-looking returns, use our forecast to generate a ranking from worst-to-best performers, and construct a relative value portfolio to capture mispricings across a six developed markets. In particular, we aim to estimate

$$\mathbb{E}\left[\frac{r^i_{t,t+h}}{\sigma^i_h}\,\middle|\,\mathcal{F}_t\right], \tag{2.1}$$

where h is the time horizon over which we aim to predict the expectation value of the returns r, and i is an index labelling one of six countries. $h = 25$ is our chosen horizon. $\mathcal{F}_t$ is the filtration at time t, and the realised vol σ^i_h is defined as the rolling standard deviation over period h of the backward-looking daily returns. We specifically work with futures on US, UK, Canadian, German, French and Italian government bonds, using historical data starting from January 2010 at the daily frequency[1]. We ensure that the features in the model are grounded in economic motivation. For instance, we use momentum in bond futures across all 6 countries as a feature, given the wealth of empirical evidence that momentum is a driver of returns in financial markets, and rates markets specifically [1–4]. We also use US treasury curve spread and fly, defined by

$$\begin{aligned}\text{Curve}_t &= y^{10\text{Y}}_t - y^{2\text{Y}}_t \\ \text{Fly}_t &= y^{2\text{Y}}_t - 2y^{5\text{Y}}_t + y^{10\text{Y}}_t,\end{aligned} \tag{2.2}$$

with y the yield of a US treasury bond, under the hypothesis that the shape of the US treasury curve reflects the market's expectations of future growth and monetary policy, and may be predictive of returns in the bond futures of other developed markets. We also use the h-day change in the MOVE index, under the hypothesis that changes in volatility regime can drive returns. Finally we define interaction terms between Curve_t and momentum; namely

$$\text{Interaction}^i_t = \text{Curve}_t \times \rho^{\lambda,i}_t, \tag{2.3}$$

where $\rho^{\lambda,i}_t$ is momentum for country i at time t, calculated with a λ-day-lookback.

Given the small cross-sectional universe and the likely multicollinearity among features, we employ ridge regression to stabilise coefficient estimates, and a rolling-window regression approach in an effort to ensure that the model coefficients adapt to market regime.

We evaluate our prediction primarily using the *information coefficient*, which we define below.

[1] We should be explicit that we make no attempt to ensure that the futures data we use is adjusted in order to be free from roll effects. The goal of this project is to provide a research prototype, rather than a production-ready strategy.

Definition 2.0.1. *Let $r^i_{t,t+h}$ be an observed return over a time period h, and let $\hat{r}^i_t$ be a prediction for that return from some model. Then the cross-sectional* ***Information Coefficient*** *$\mathcal{I}_t$ is defined*

$$\mathcal{I}_t := \mathbb{C}\mathrm{orr}_i \left(\hat{r}^i_{t,t+h}, r^i_{t,t+h}\right), \tag{2.4}$$

the correlation between the predicted and realised returns.

Since we predict volatility-scaled returns in this work, the information coefficient we use is

$$\mathcal{I}_t := \mathbb{C}\mathrm{orr}_i \left(\hat{y}^i_{t,t+h}, y^i_{t,t+h}\right), \tag{2.5}$$

where we define

$$\hat{y}^i_{t,t+h} := \frac{\hat{r}^i_{t,t+h}}{\sigma^i_h}, \quad y^i_{t,t+h} := \frac{r^i_{t,t+h}}{\sigma^i_h}. \tag{2.6}$$

$\mathbb{C}\mathrm{orr}_i$ is the Pearson correlation across assets, namely

$$\mathbb{C}\mathrm{orr}_i \left(\hat{y}^i_{t,t+h}, y^i_{t,t+h}\right) := \frac{\mathbb{C}\mathrm{ov}_i \left(\hat{y}^i_{t,t+h}, y^i_{t,t+h}\right)}{\sqrt{\mathbb{V}\mathrm{ar}_i \left(\hat{y}^i_{t,t+h}\right) \mathbb{V}\mathrm{ar}_i \left(y^i_{t+h}\right)}}, \tag{2.7}$$

with

$$\mathbb{C}\mathrm{ov}_i \left(\hat{y}^i_{t,t+h}, y^i_{t,t+h}\right) := \frac{1}{N} \sum_{i=1}^{N} (\hat{y}^i_{t,t+h} - \bar{y}_{t,t+h})(y_{t,t+h} - \bar{y}^i_{t,t+h}), \tag{2.8}$$

$$\mathbb{V}\mathrm{ar}_i \left(y^i_{t,t+h}\right) := \frac{1}{N} \sum_{i=1}^{N} (y_{t,t+h} - \bar{y}^i_{t,t+h})^2, \tag{2.9}$$

and similar for $\mathbb{V}\mathrm{ar}\left(\hat{y}^i_{t,t+h}\right)$.

3 Data & Feature Engineering

3.1 Data

We use 10Y Government Bond futures across 6 major developed markets, the MOVE index[2], and 2-year, 5-year & 10-year US Treasury yields. All data is sourced from `investing.com` and sampled at the daily frequency, with the raw dataset spanning January 2010 to March 2026.

3.2 Return and Volatility Definitions

We define the h-day forward-looking log returns as

$$r^i_{t,t+h} := \log\left(\frac{P^i_{t+h}}{P^i_t}\right), \tag{3.1}$$

and the realised volatility as the h-day rolling standard deviation of the backward-looking daily log returns. Namely, we let

$$r^i_t = \log\left(\frac{P^i_t}{P^i_{t-1}}\right), \tag{3.2}$$

and define

$$\sigma^i_{h,t} := \sqrt{252} \cdot \sqrt{\frac{1}{h}\sum_{j=t-h}^{t-1}\left(r^i_j - \bar{r}^i_{t-1,h}\right)^2}, \tag{3.3}$$

with

$$\bar{r}^i_{t-1,h} := \frac{1}{h}\sum_{j=t-h}^{t-1} r^i_j. \tag{3.4}$$

3.3 Target Construction

The target is the vol-scaled h-day forward-looking log return,

$$y^i_{t,t+h} := \frac{r^i_{t,t+h}}{\sigma^i_t}. \tag{3.5}$$

From an investment perspective, we are more interested in delivering stable, *risk-adjusted* returns rather than simply trying to maximise raw returns. Thus, we define the regression target to be the vol-scaled return as opposed to the raw return. This ensures that the target is aligned with our investment objective, and also allows assets of differing volatility levels to be compared. In this way, the ranking of assets we build during the portfolio construction is a ranking in expected return per unit of risk[3], rather than one based on absolute return magnitude.

[2] MOVE is the Merrill Lynch Option Volatility Estimate. This is sometimes referred to as the "VIX for bonds"; the VIX provides an estimate of volatility in equity markets, whereas the MOVE index instead describes volatility in the US Treasury market.

[3] We implicitly use realised volatility as a proxy for risk throughout.

Prediction Horizon. The prediction horizon h is set to 25; namely for each trading day t we aim to predict the risk-adjusted return h days ahead.

3.4 Feature Set

Momentum We define momentum for asset i at time t as the λ-day backward-looking log return,

$$\rho_t^{\lambda,i} := \log\left(\frac{P_t^i}{P_{t-\lambda}^i}\right). \tag{3.6}$$

Our motivation for using momentum as a feature is the widely documented phenomenon that momentum has predictive value in financial markets, and rates markets in particular. Trends in bond markets are often attributed to slow dissemination of macro information and the subsequent gradual rebalancing by market participants.

In our case, we set the lookback λ to 50 days - just over 2 months' worth of trading days.

Curve Shape As stated in the overview, we use two curve-based features capturing the shape of the US treasury curve;

$$\begin{aligned} \text{Curve}_t &= y_t^{10\text{Y}} - y_t^{2\text{Y}} \\ \text{Fly}_t &= y_t^{2\text{Y}} - 2y_t^{5\text{Y}} + y_t^{10\text{Y}}. \end{aligned} \tag{3.7}$$

The economic rationale for these features is that the shape of the yield curve implies the market's expectations for future macro regime; namely growth, inflation and monetary policy expectations. The hypothesis is that such expectations reflected in the US treasury curve are incorporated into prices across markets at different speeds, leading to relative mispricings that can potentially be exploited.

Volatility Regime We also use the h-day change in the MOVE index as a feature, in an effort to capture the effect of shifts in volatility regime on bond future returns. Changes in volatility regime can lead investors to rebalance their portfolios, with increases in volatility often the cause of increased risk aversion.

Interactions Finally, we use a number of interaction terms between US treasury curve spread and momentum for each country in the cross-section. Namely,

$$\text{Interaction}_t^i := \text{Curve}_t \times \rho_t^{\lambda,i} \tag{3.8}$$

These interaction features attempt to capture the effect of momentum driving returns differently depending on curve shape. The rationale is that trend-following behaviour by market participants may be exacerbated or weakened by, for example, monetary policy regime. The interaction features allow the model to capture such effects.

3.5 Panel Construction

We stack the data detailed in section 3.1, into a single panel dataset with one row per pair of (date, asset). Namely, after ordering the dataset by date, each entry in the date column is repeated 6 times; once for each of the 6 countries.

We could opt to perform 6 separate regressions rather than a single panel regression. However, stacking the assets into a panel allows us to train a single model on all assets simultaneously, which has a number of benefits:

1. It gives the model more data to learn from, improving coefficient stability,
2. It allows the model to capture cross-asset effects, rather than having 6 separate models that are each blind to the others,
3. It aligns with our investment objective; namely we can predict the entire cross-section at once and construct a ranking across assets.

4 Modelling Approach

The central modelling approach is the following.

> We perform a rolling-window ridge regression on our panel of assets, predicting risk-adjusted returns for each asset.

Given that the features are likely to exhibit multicollinearity, ordinary least-squares regression is likely to be unstable; small changes in the input data are likely to produce large changes in the learned coefficients. Regularization methods can mitigate multicollinearity in the features and improve coefficient stability - we choose ridge regression over the Lasso regression given that each of our features are grounded in an economic rationale and we therefore suspect that they all have some degree of predictive power[4] .

We specifically perform a *rolling* ridge regression. That is, for each day t in the dataset, we:

1. Look back a specified number of days, which we call `rolling_window`,
2. Train the model on the last rolling_window's worth of days,
3. Predict that day's cross-section,
4. Move forward one day,
5. Repeat.

[4]Ridge regression, unlike lasso regression, does not perform variable selection, and so will not discard any features from the training set.

The reason we do this is so that the model coefficients adapt to market regime - if we just trained on the entire dataset we'd be treating market conditions as identical across time, which is clearly not correct. For example, April 2025 (tariffs) does not represent the same market conditions as early 2020 (COVID selloff), which does not represent the same market conditions as the post-COVID hiking cycle.

Window length. The window length `rolling_window` is set to 252; approximately one year's worth of trading days. The rationale is that this is enough data that the model shouldn't overfit to noise, but the window is small enough such that the model is trained on current market regimes. Another way to word the latter is that we are assuming that for a window length of 252 the relationship between the features and the target is roughly homogenous across the window.

Standardisation. Before regression, all features are standardised within each rolling window - each column of the feature matrix X is de-meaned and scaled by its own standard deviation[5] - in order to ensure that assets are comparable in scale such that the regularisation penalty treats all coefficients equally.

Hyperparameter. The only hyperparameter in ridge regression is the value of the penalty parameter, referred to in the appendix as λ. This is chosen to be 1 in the model, but a more thorough analysis would choose the optimal parameter with a systematic approach.

Portfolio Construction Predicted risk-adjusted returns are converted into cross-sectional weights by first de-meaning the predictions, before scaling each by the sum of the absolute values of each asset's de-meaned prediction. In particular, for each prediction p_t^i, we compute $x_t^i = p_t^i - \bar{p}_t$ where $\bar{p}_t = \frac{1}{6}\sum_{i=1}^{6} p_t^i$ and define the portfolio weights to be

$$w_t^i := \frac{x_t^i}{\sum_{i=1}^{6} |x_t^i|}. \tag{4.1}$$

Finally, we shift the weights by 1 to prevent lookahead bias. Namely, we want to ensure that the portfolio we will trade at time t uses weights based on predictions made from information available up to time $t-1$, and no future data.

We also note that the portfolio is rebalanced daily. This leads to high turnover, and as we will see this impacts the results when transaction costs are taken into account.

[5]This results in each feature having mean 0 and standard deviation 1. Intuitively, we standardise over assets (taking the mean and standard deviation over assets at fixed time and scaling appropriately) because we want to be able to compare different assets whose features might have different levels on the same day, and we don't want the model to simply learn levels. We also standardise over time within the rolling window because it doesn't make sense to only standardise on one fixed day - we want to account for changes in levels over time. A momentum of x at time t needs to have a meaning relative to its historical context in time, rather than just to the other assets at t.

5 Strategy Performance

We evaluate the strategy's performance by looking at its information coefficient, turnover, returns and Sharpe ratio. We also investigate the strategy's orthogonality to a basic risk factor - is the strategy simply momentum-riding in disguise?

Information Coefficient. The information coefficient[6] captures the correlation between the model's predictions and the corresponding observation. The information coefficient is bounded between 1 and -1, with 0 indicating that the model has no predictive power and -1 indicating that the prediction is inversely correlated with the realised return. We see that the information coefficient swings

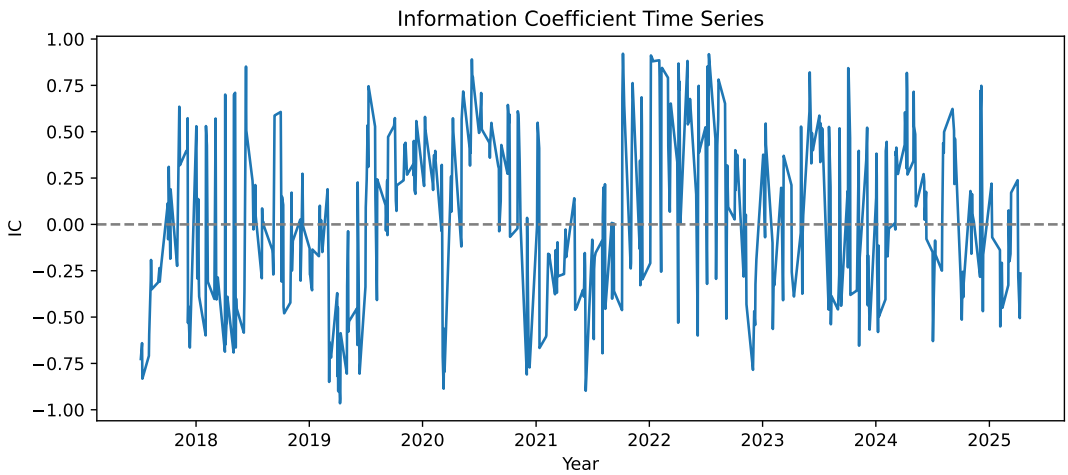

Figure 1: A plot showing the information coefficient as a function of time.

rather rapidly between large positive and large negative values. This could be an indication that the model has some degree of predictive power and may potentially generate some alpha, but that the signal is unstable and may be hard to reliably monetise. We should interpret such IC values with caution - given our rather small cross-section of only 6 assets, correlation estimates are inherently noisy and so large IC values do not necessarily imply strong predictive power. It is interesting to note that the IC spends a lot of time near 1 during 2022, which coincides with a steep gain in the strategy during this period - see the cumulative return plots later in this report.

It is also informative to plot the distribution of the information coefficient. We see that the distribution is rather broad, as expected from the time series plot. In this setting, the high variability in the IC is likely to be largely driven by estimation noise due to the small cross-section, though it may also indicate that any underlying signal is weak or regime-dependent.

The mean IC is ~ 0.049, with a standard deviation of ~ 0.439. This gives a t-statistic of ~ 2.525. The positive mean indicates that the model could have some degree of predictive power, though

[6]Note that the IC we use here is specifically the *Pearson* IC - there exist other definitions of IC based on other correlation measures.

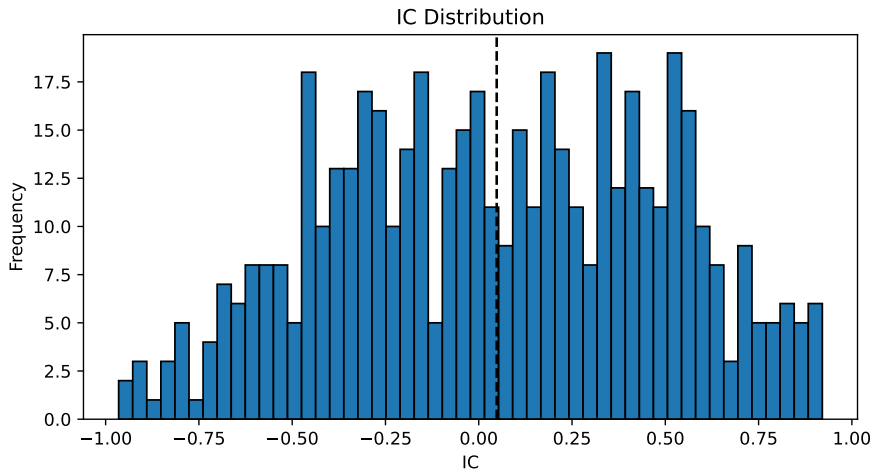

Figure 2: A plot showing the distribution of the information coefficient. We see that the distribution is rather broad; one way to understand this is that our cross-sectional universe is rather small - estimating the Pearson correlation between two variables with only 6 data points each is subject to instability.

this should be interpreted with caution for the reasons mentioned previously. In a similar vein, the relatively large standard deviation highlights that the signal is highly variable over time, consistent with the noisy behaviour observed in figures 1 and 2. The resulting t-statistic indicates that the mean IC is statistically different from zero at conventional significance levels, though again this should be interpreted with caution given the small cross-sectional sample used to compute each IC observation.

Overall, these results suggest that while the signal may contain some genuine predictive power, it is weak and unstable, and monetisation of the strategy as-is would likely pose a challenge. This brings us to the issue of turnover.

Turnover. Our per-asset contribution to the turnover is defined by

$$\mathcal{T}_t^i := |w_t^i - w_{t-1}^i|, \tag{5.1}$$

with the aggregate daily portfolio turnover defined by a sum over assets,

$$\mathcal{T}_t := \sum_{i=1}^{6} \mathcal{T}_t^i. \tag{5.2}$$

This implies a daily rebalancing strategy that we discuss the implications of in more detail in later sections. We see in figure 3 that the turnover is highly variable, with frequent rebalances. The turnover is regularly close to 1, indicating a substantial reshuffling of the portfolio, with frequent spikes up to 2 - a total reversal of portfolio positions[7]. Such frequent trading with even modest

[7] A total reversal of positions occurs when $w_t^i = -w_{t-1}^i$, which gives $\mathcal{T}_t = 2\sum_i |w_{t-1}^i| = 2$.

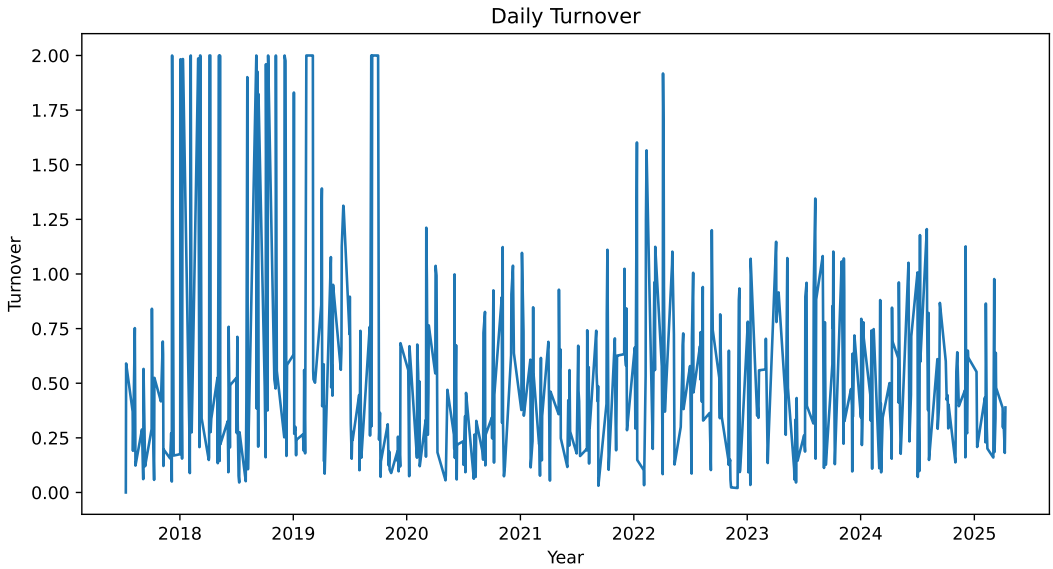

Figure 3: A plot showing the portfolio turnover; namely the absolute change in portfolio weights as a function of time. In the naive daily-rebalancing scheme used in this project, transaction costs completely erode the gross alpha.

transaction costs is likely to completely erode gross alpha if the signal is weak, as we suspect it is given our analysis of the information coefficient. In addition, the high turnover implies that the model could be capturing short-term dislocations, rather than structural trends.

Strategy Returns. The strategy's cumulative returns with and without transaction costs are shown in figure 4. The strategy shows a steep gain in the first half of 2022, namely in a macro regime characterised by inflation fears sparked by the war in Ukraine and subsequent aggressive rate hikes. The latter half of 2022 shows the strategy experiencing greater-than-average volatility, and cumulative returns seem to level off after 2024.

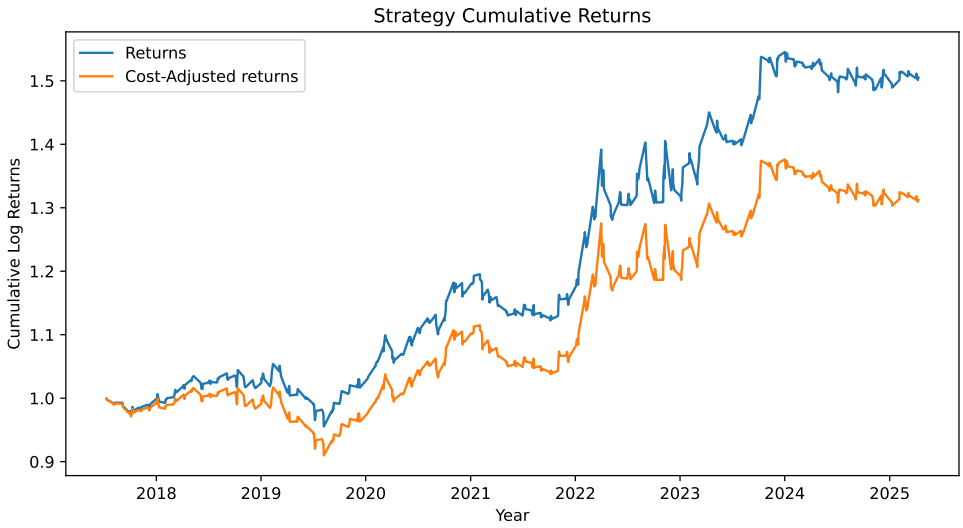

Figure 4: The cumulative returns of the strategy as a function of time, both with and without transaction costs of 5bps per trade.

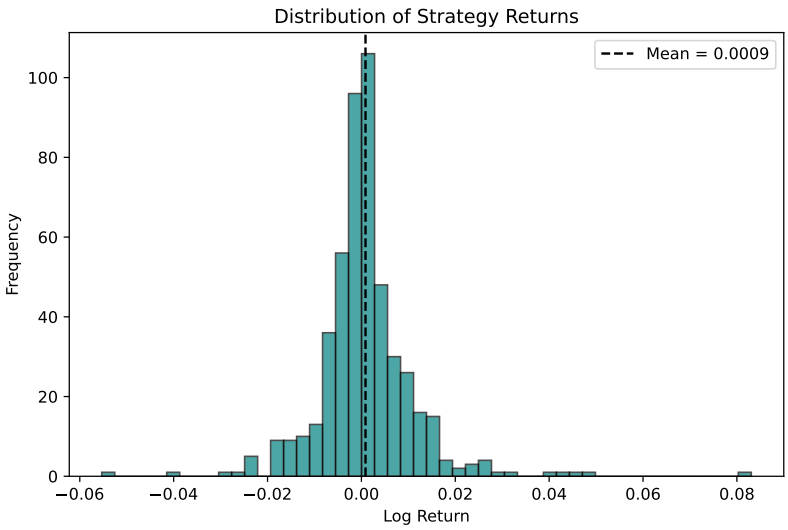

Figure 5: The distribution of the strategy's returns, *without* transaction costs.

Sharpe & Orthogonality The strategy produces the following statistics:

- Sharpe ratio: ~ 1.34 without transaction costs, ~ 0.92 with costs.

- Max Drawdown: $\sim -9.35\%$.

One might question these Sharpe ratios given that we are predicting h-day forward returns, but rebalancing the portfolio every day. Because in the ridge regression loop the prediction windows overlap, the daily returns for the days in the window that overlap are correlated, lowering the daily variance of the strategy. In the presence of a strong signal, the daily rebalancing strategy could serve to systematically diversify away our entry-timing risk. The calculated Sharpe ratio then reflects the risk-adjusted performance of this daily-rebalancing execution strategy, as opposed to being an artificial inflation of the Sharpe. However, as we shall see below, the strategy's high turnover turns out to erode the weak signal.

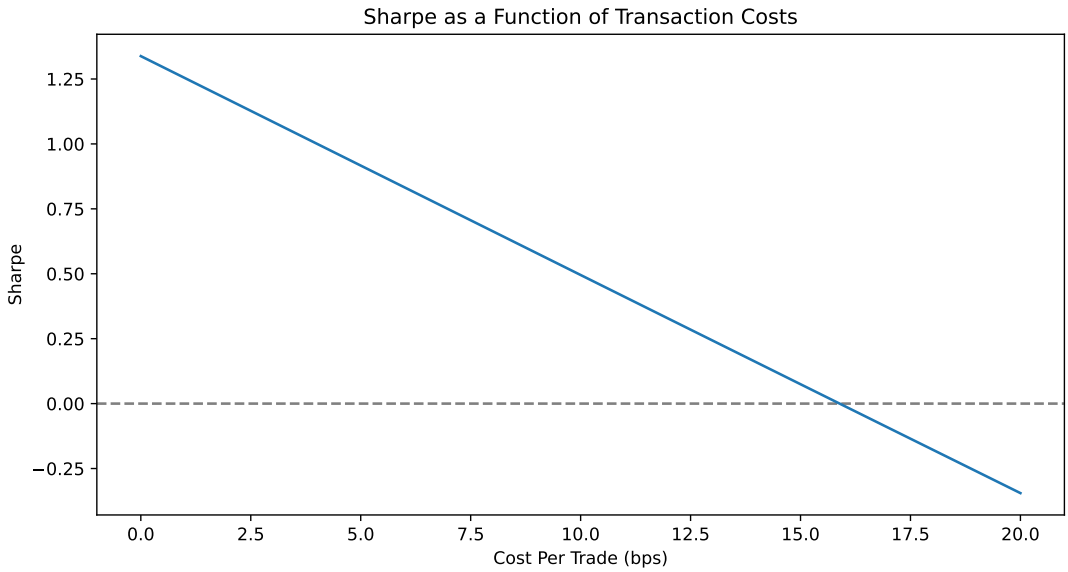

Figure 6: The Sharpe ratio decreases linearly with trading costs in a naive linear cost model that ignores market impact and slippage. The Sharpe hits zero at a cost of roughly 15.9 bps per trade.

One question that is important to ask is - does the strategy deliver returns that are orthogonal to basic momentum? We investigate this by performing two basic OLS regressions - we regress the strategy returns against momentum with and without transaction costs. We find the following regression results:

- Without transaction costs:
  - α: ~ 0.00016 (~ 1.6bps)
  - β: ~ -0.0015
  - R^2: 0.016.
- With transaction costs:

- α: ~ -0.00096 (~ -9.6bps)
- β: ~ -0.0016
- R^2: 0.017.

The low R^2 values and near-zero β coefficients across both regressions indicate that the strategy's returns are orthogonal to a simple momentum factor[8]. This suggests that any predictive power of our model is driven by the structural features (US Treasury curve spread, fly, and interaction terms) rather than the strategy being simply momentum-riding in disguise. However, the regression intercept, describing the daily residual return independent of momentum, indicates that the signal is weak and eroded by our naive execution strategy. Before transaction costs, the strategy generates a positive residual return of approximately 1.6bps per day. Once a 5-bps-per-trade cost is introduced, this daily residual return drops to approximately -9.6 bps.

It is interesting to note that despite the fact that the regression alpha is negative after incorporating transaction costs, the strategy is still profitable. In our high-turnover, daily-rebalancing strategy, the regression alpha represents the average daily return above simple momentum, whereas the cumulative PnL sums the actual daily returns over time. Transaction costs reduce the daily mean return, causing the regression alpha to flip negative, even though the strategy occasionally experiences large gains that dominate the cumulative returns. These intermittent positive swings can outweigh the small negative daily mean, resulting in a positive cumulative PnL despite a negative regression alpha.

[8]In this context the R^2 value describes how much of the variance in the strategy's returns are explained by basic momentum, and the β describes the sensitivity of the strategy's returns to momentum.

6 Discussion

6.1 Short Summary

This project developed a systematic, cross-sectional, relative value trading strategy on 10-year Government Bond futures across 6 developed markets. The central research hypothesis was that relative performance across global bond markets is partly driven by momentum, changes in volatility regime and US Treasury curve structure, and that such signals are incorporated into prices at different speeds across markets, leading to relative mispricings that can be exploited.

We found that the strategy produces a Sharpe of ~ 0.92 and ~ 1.34 with and without transaction costs of 5bps per trade respectively, and a maximum drawdown of $\sim -9.35\%$. We found that the Sharpe hits 0 at a cost-per-trade of ~ 15.9bps. We found that the strategy's returns are orthogonal to basic momentum and that residual returns above momentum are approximately 1.6bps per day. However, these residual returns are completely eroded upon introducing transaction costs due to our execution strategy, where the portfolio is rebalanced daily. This erosion is consistent with the high turnover of the strategy, which frequently approaches a full reversal of positions ($\mathcal{T}_t \sim 2$). The observed information coefficient was highly variable over time with a broad distribution, reflecting the small cross-section of 6 assets and suggesting a weak, potentially regime-dependent predictive signal.

6.2 Directions For Future Work

There are a number of directions for future research that would be interesting to pursue.

- **Execution.** The results suggest that the "raw" signal may be valid (if weak), but the current implementation suffers from over-trading due to the daily rebalancing, eroding the residual returns above basic momentum. It would be interesting to investigate whether or not the signal can survive transaction costs in a less aggressive execution strategy, perhaps trading only when the portfolio weights cross a certain threshold.
- **Trading Frictions.** While we gave a basic account of the strategy's performance in the presence of transaction costs, we did not attempt to model slippage, market impact, or non-linear trading costs. Modelling such effects in conjunction with the above investigation into execution strategy is a natural next step for this work.
- **Model Approach.** Since we dealt with features that were likely to be highly multicollinear, ridge regression was a natural model choice. However, the Tikhonov parameter was simply set to 1 from the outset, whereas ideally one would perform some hyperparameter tuning to find the optimal value. In addition, other regularisation methods are available - the Lasso is a common approach, which performs both regularisation and variable selection. We discarded the Lasso since we wanted to retain all features, but it would be interesting to see whether variable selection improves the predictive power of the signal.
- **Futures Roll Handling.** As futures contracts approach expiry, the price of the contract approaches the spot price of the underlying. The new front-month contract will then in

general have a different price from the expiring contract, reflecting the cost-of-carry. This can lead to jumps in the price series at contract roll dates. Since the project was intended as a research prototype rather than an effort to build a production-ready strategy, we did not explicitly adjust for roll effects. In a professional research pipeline we would ensure that the data is either adjusted for contract rolls in the first place, or in the case that it is not we would construct a continuous futures series to ensure that returns used in the model are driven by price dynamics, rather than contract transitions.

A Regression Methods

The appendix serves as a short introduction to linear regression, and ridge regression as a method for guarding against the problems that multicollinearity can cause.

A.1 OLS Regression

In Ordinary Least Squares regression, we fit a linear function f to input data $X \in \mathbb{R}^{n \times p}$ and output data $Y \in \mathbb{R}^n$, assuming the relationship between inputs and outputs is of the form

$$Y = X\beta + \varepsilon, \tag{A.1}$$

where $\beta \in \mathbb{R}^p$ and ε is a vector of *errors* or *residuals*, capturing random noise in Y that is not explained by the model. OLS regression makes a number of assumptions about the data, which we delineate below.

Linearity. We assume that the relationship between the *feature matrix* X and the *target* Y is linear, of the form (A.1).

Homoscedasticity. We assume homoscedastic errors, namely that the variance of the residuals ε is constant across all values of inputs. We make this assumption because the opposite, heteroscedasticity, can lead to unreliable coefficient estimates.

To see this, consider that by definition, in the presence of heteroscedasticity some groups of datapoints are much more widely dispersed than others. Given that the job of regression is to estimate a straight line that passes as close to all of the datapoints as possible, the points with largest variance will pull the regression line towards themselves in order to minimise the overall scatter of the datapoints about the regression line.

We can overcome heteroscedasticity by weighting each of the datapoints by the inverse of their variance. However, this assumes that we have access to the variance associated with each datapoint, which often we do not.

Normality. Errors are normally distributed with mean 0 and variance σ^2, namely $\varepsilon \sim \mathcal{N}\left(0, \sigma^2\right)$.

Normality of the residuals implies that the OLS estimate of the regression coefficients are distributed as

$$\hat{\beta} \sim \mathcal{N}\left(\beta, (X^T X)^{-1}\sigma^2\right), \tag{A.2}$$

a fact which is in turn used to form confidence intervals and hypothesis tests for the regression coefficients β_j.

Independence. Errors are assumed to be independent. OLS does not require this assumption to produce unbiased coefficients, but again this assumption is needed for valid confidence intervals and hypothesis tests.

An example of a potential problem that can arise from violation of independence is that if errors are correlated, they tend to occur in "runs" - for example if the error ε_j for datapoint Y_j is large, and ε_j and ε_k are correlated, then ε_k may be likely to be large too. Prediction intervals (the range in which a new datapoint is likely to fall) then become too narrow, since OLS regression assumes independence and does not account for "momentum" in the errors.

Lack of Multicollinearity. Most pertinent to this report, *Multicollinearity* is when two or more features are highly correlated with eachother[9]. Multicollinearity is a problem for two reasons:

1. Regression coefficients become unstable - small changes in input data produce large changes in the coefficients.
2. Interpretability becomes more difficult - for two features X_1 and X_2 it is difficult to make conclusions like "X_1 increases Y by ...", if X_1 is correlated with X_2, since it is challenging to separate their contributions.

To make this more precise, OLS regression minimises the *Residual Sum-of-Squares*,

$$\mathrm{RSS}(\beta) := (Y - X\beta)^T (Y - X\beta), \tag{A.3}$$

which is minimised by

$$\hat{\beta}_{\mathrm{OLS}} = (X^T X)^{-1} X^T Y. \tag{A.4}$$

Features are said to be multicollinear when

$$X_j \approx c_1 X_k + c_2 X_\ell + \ldots, \quad j \neq k \neq \ell \tag{A.5}$$

for X_i columns of the feature matrix X. If it is the case that one or more of the columns of X is nearly a linear combination of other columns, then the matrix $X^T X$ will be nearly singular, and some elements of its inverse $(X^T X)^{-1}$ will be very large. The variance of $\hat{\beta}_{\mathrm{OLS}}$ is given by

$$\mathbb{V}\mathrm{ar}[\hat{\beta}_{\mathrm{OLS}}] = \sigma^2 (X^T X)^{-1}, \tag{A.6}$$

and so explodes if $X^T X$ is nearly singular. Another way to see that $\hat{\beta}_{\mathrm{OLS}}$ becomes unstable is by noting that any small perturbation of $X^T Y$ (namely, the input data changing slightly) will be amplified by the large elements of $(X^T X)^{-1}$ in (A.4).

[9] *Perfect* collinearity is when one feature is an exact linear combination of others.

In this project, it is likely that our features exhibit a high degree of multicollinarity. Therefore, ordinary least-squares is not a sensible choice of model, and ridge regression is chosen instead to mitigate the effects of multicollinearity.

A.2 Ridge Regression

Ridge regression is a common technique for stabilising regression coefficients in the presence of multicollinearity. In ridge regression, rather than minimising the residual sum-of-squares (A.3), we instead minimise

$$\mathcal{R}(\beta) := (Y - X\beta)^T(Y - X\beta) + \lambda\beta^T\beta, \tag{A.7}$$

with $\lambda \in \mathbb{R}$ sometimes called the *Tikhonov parameter*. The "penalty" term $\lambda\beta^T\beta$ penalises large regression coefficients. The ridge regression coefficients are given by

$$\hat{\beta}_{\text{Ridge}} = (X^TX + \lambda\mathbb{I})^{-1}X^TY, \tag{A.8}$$

namely a positive constant is added to X^TX before inversion. Therefore, in the presence of multicollinearity when X^TX is nearly singular, the effects mentioned earlier are mitigated by the fact that $X^TX + \lambda\mathbb{I}$ can be tuned away from singularity.

One way to see this is by a quick argument about the eigenvalues of X^TX - if X^TX is nearly singular, then $\det(X^TX) \approx 0$, which is trivially equivalent to $\det(X^TX - 0 \cdot \mathbb{I}) \approx 0$, so X^TX has a near-zero eigenvalue. If X^TX has eigenvalues λ_k, then $(X^TX + \lambda\mathbb{I})$ has eigenvalues $\rho_k = \lambda_k + \lambda$, which are never close to zero if λ is sufficiently positive, and so $(X^TX + \lambda\mathbb{I})$ is not nearly singular.

Since $X^TX + \lambda\mathbb{I}$ can be tuned away from singularity, small perturbations of the input data will no longer be amplified by its inverse, stabilising the regression coefficients $\hat{\beta}_{\text{Ridge}}$.

References

[1] Tobias J. Moskowitz, Yao Hua Ooi, Lasse Heje Pedersen, Time series momentum, Journal of Financial Economics, Volume 104, Issue 2, 2012, Pages 228-250, ISSN 0304-405X, https://doi.org/10.1016/j.jfineco.2011.11.003.

[2] Baz, Jamil and Granger, Nicolas and Harvey, Campbell R. and Le Roux, Nicolas and Rattray, Sandy, Dissecting Investment Strategies in the Cross Section and Time Series (December 4, 2015). Available at SSRN: https://ssrn.com/abstract=2695101 or http://dx.doi.org/10.2139/ssrn.2695101

[3] Hurst, Brian and Ooi, Yao Hua and Pedersen, Lasse Heje, A Century of Evidence on Trend-Following Investing (June 27, 2017). Available at SSRN: https://ssrn.com/abstract=2993026 or http://dx.doi.org/10.2139/ssrn.2993026

[4] Jegadeesh, N. and Titman, N, S. (1993), Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. The Journal of Finance, 48: 65-91. https://doi.org/10.1111/j.1540-6261.1993.tb04702.x